SENTIENT FUTURES FELLOWSHIP · RESEARCH REPORT

Protein futures in India and Pakistan

Drivers, the role of AI, and speculative market scenarios to 2035, with talent and policy implications

Author Muhammad Umar Zafar

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Status AI-assisted working report. Human scholarly review pending. The market figures are speculative scenarios, not forecasts or observed sales.

Companion Interactive research edition with all tables and sources: pakistan-india-protein.vercel.app

1. Summary

India and Pakistan could develop very different alternative-protein markets by 2035, yet the evidence cannot yet say how large either will be. Middle scenarios put annual spending at about $448 million in India and $11.8 million in Pakistan, both far below US spending per person.

$448 M
India 2035 middle scenario (range $166 M to $1,142 M), constant 2025 dollars

$11.8 M
Pakistan 2035 middle scenario (range $3.6 M to $36.1 M)

1.22% / 0.17%
India / Pakistan spending per person in 2035 as a share of US spending in 2025

0 of 9
published consumer studies that measure observed purchase in either country

The report asks which drivers will shape alternative-protein demand in India and Pakistan over 10 years, what AI could change, and where talent and funding should go first. It reaches seven findings.

1. **Supply differs sharply.** In 2023 Pakistan had 8.72 kg of poultry available per person against India's 3.52 kg. India had 15.39 kg of pulses against Pakistan's 6.53 kg. Total protein supply is nearly equal at 74.6 and 73.4 g per person per day.
2. **Affordability is the sharper contrast.** A healthy diet was out of reach for 60.3% of Pakistan's population and 40.4% of India's in 2024.
3. **Market scenarios are small and wide.** The 2035 India range spans a factor of 7. The model sits below GFI India's lowest 2030 scenario in every one of its 10,000 draws, and the gap has no single explanation yet.
4. **An AI forecast panel disagrees with itself.** Two models, asked 10 times each for India, put its 2035 market at about 370 and 158 million US dollars, constant 2025 prices. Their 10th to 90th percentile ranges of run medians never overlap. Section 5.7
5. **AI is a plausible driver with no isolated evidence.** Experimental work supports AI in culture-media search and protein design. No source tested here shows a cost or yield effect in either country.
6. **Talent signals favour India.** India posted 529 vacancies on one job board in 2025 and Pakistan 58. These are demand signals, not shortages.
7. **Behaviour is unmeasured.** Willingness, first purchase, repeat purchase and substitution of animal foods have no observed baseline in either country.

> **How to read the numbers.** Proxy market scenarios use analyst-chosen inputs without calibration against verified sales. The AI panel compares structured forecasts from two models. Neither method supplies observed sales. Treat both as ways to compare assumptions. A verified sales total would improve the work more than further refinement.

Recommendations in brief

- Fund employer-linked technical trials in India and a small ingredient-application pilot in Pakistan before expanding training programmes.
- Require independently checked AI versus non-AI yield and cost-per-protein data before adding any AI effect to a market scenario.
- Run a purchase-tracking study that links stated intention to verified transactions in a defined regional cohort.
- Shared definitions for currency, price year, value basis and product scope were agreed with the US and Africa authors on 22 October 2026.

2. Question, scope and evidence rules

The Sentient Futures mentees are asking how AI could reshape the food system transition over 10 to 20 years and where talent and resources should go now. This report covers India and Pakistan. A US section and an Africa section, written by other mentees, will sit beside it. The three sections share one question: how large will the alternative-protein market be in 2026, 2030 and 2035?

What counts as alternative protein here

The scope is domestic retail sales of plant-based meat, dairy and egg alternatives and fermentation-derived foods for human consumption. Cultivated meat enters only after an authorisation exists. The US and Africa authors agreed this scope on 22 October 2026.

Method in five steps

1. **Describe the food system.** National food balances (FAOSTAT, release of 28 October 2025), household surveys in each country, and the 2024 nutrition indicators give the baseline.
2. **Map the drivers.** Seven drivers, plus AI, are traced from evidence to a direction for 2026, 2030 and 2035, each with a signpost that would confirm or reject the direction.
3. **Build speculative scenarios.** A Monte Carlo model (10,000 draws, seed 20261015) anchors India to a 2022 GFI India estimate and scales it by income, urban share and annual drift. Pakistan scales from India by relative income, urban share, animal-protein supply and industry activity.
4. **Compare repeated AI forecasts.** Adapt Anna Bartlett's questionnaire and compare 10 fresh-context answers per model and region. Report medians and 10th to 90th percentiles of run medians, keeping proxy scenarios separate. Section 5.7 describes all 60 accepted runs.
5. **Test the limits.** The model is compared with GFI India's own 2030 scenarios, a drift-overlap sensitivity, and the held-out interval coverage of the supply benchmark.

Five evidence rules

- Food supply is not individual intake. A food balance sheet records national availability.

- Willingness to try, purchase, repeat purchase and substitution are four different outcomes.
- Consumer studies from different countries and products are not pooled into one acceptance rate.
- Approval of a food, halal certification and market uptake are three separate events.
- A missing baseline stays missing. No scenario midpoint fills it.

Evidence status

The data package was rebuilt and checked on 20 October 2026: 25 check groups pass, including 31,575 library checks. Those checks test the software and the source joins. They do not replace a qualified reader's judgement. No interviews, tasting trials or consumer surveys were run for this report, and no verified national alternative-protein sales figure was found. Section 5.7 adds the retained AI forecast panel.

3. The driver map

Seven drivers could move demand. Each one below carries the strongest current evidence, a speculative direction by year, and a signpost. The directions are hypotheses about decisions. They are not estimated causal effects.

3.1 Two food systems, similar total protein

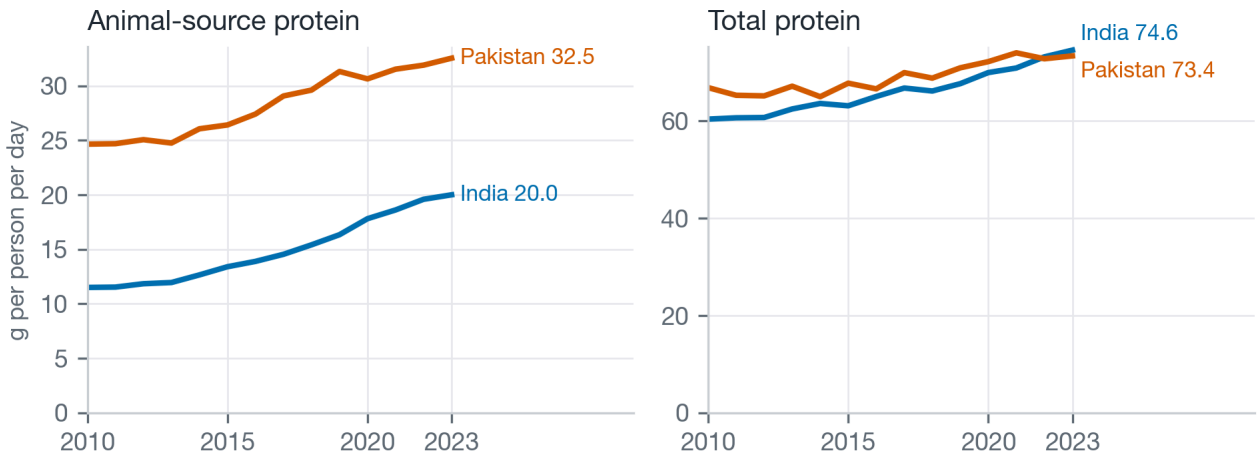


Figure 1. Pakistan supplies more animal-source protein; total protein is about equal. In 2023 animal-source supply was 20.0 g per person per day in India and 32.6 g in Pakistan. Total supply was 74.6 g and 73.4 g. Source: FAOSTAT food balances, release 28 October 2025. National supply, not individual intake.

India closed most of the gap in total protein between 2010 and 2023, rising from 60.4 to 74.6 g per person per day. Pakistan rose from 66.9 to 73.4 g. The mix differs. Pakistan draws 44% of its protein supply from animal sources (32.6 of 73.4 g) and India 27% (20.0 of 74.6 g). A new protein that replaces animal foods therefore has more to replace in Pakistan and more familiar plant foods to compete with in India.

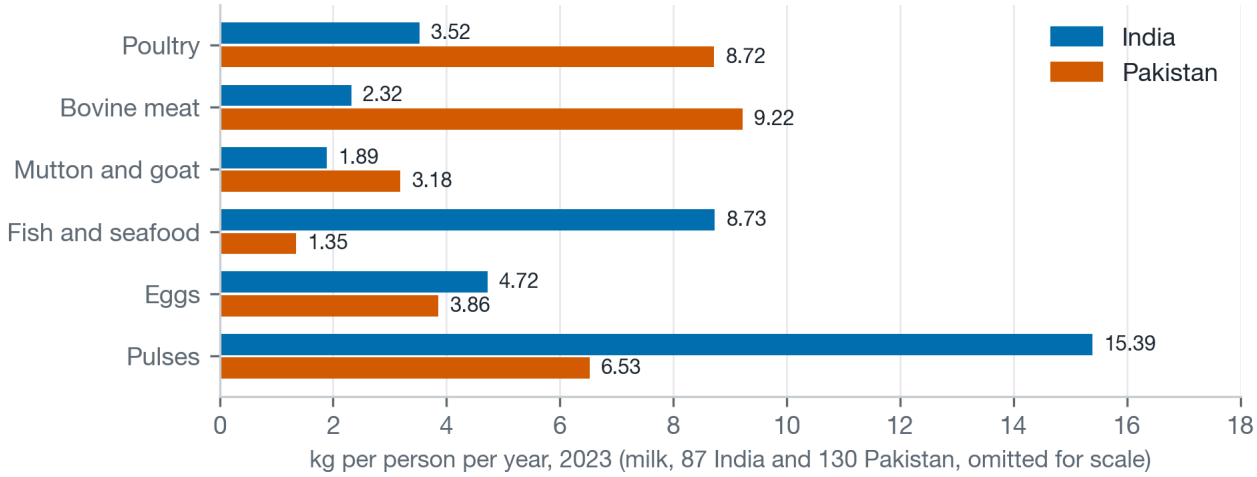


Figure 2. The two countries rely on different foods. Pakistan supplies 2.5 times India's poultry and 4 times its bovine meat. India supplies 2.4 times Pakistan's pulses and 6.5 times its fish and seafood. Source: FAOSTAT food balances, 2023.

3.2 Access, not only preference

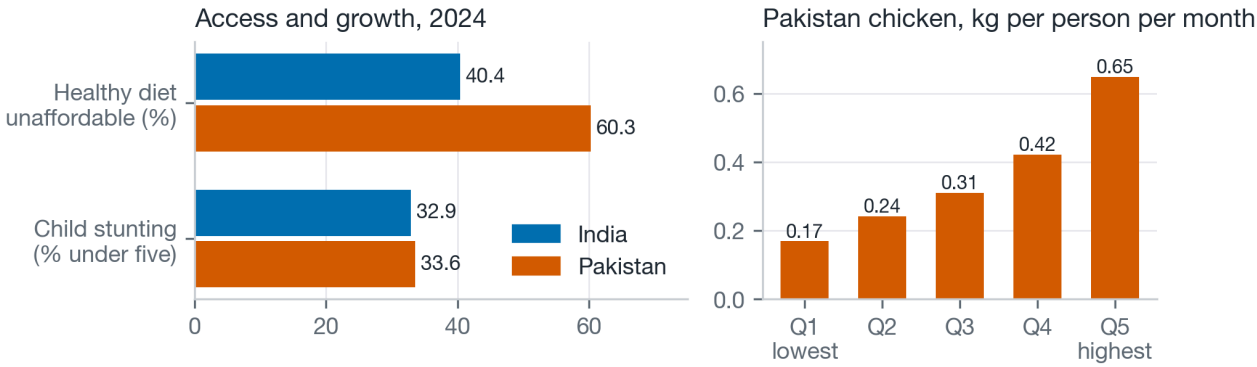


Figure 3. Pakistan's affordability gap is wider than its stunting gap, and chicken buying rises with spending. Left: healthy-diet unaffordability is 40.4% in India and 60.3% in Pakistan; child stunting is 32.9% and 33.6%. The measures use different denominators and must not be added. Right: published chicken quantities by expenditure quintile, Pakistan HIES 2024 to 2025. Sources: SOFI 2025, JME 2026, PBS.

Household surveys show the same pattern inside Pakistan. Chicken quantities in the 2024 to 2025 Household Integrated Economic Survey rise from 0.169 kg per person per month in the lowest expenditure quintile to 0.648 kg in the highest, a factor of 3.8. A product priced above chicken will meet that gradient. Its first buyers will come from the upper quintiles.

3.3 Driver table

Table 1. Seven drivers, with evidence, speculative direction and signpost

Driver	Strongest current evidence	Direction (speculative)	Signpost
Income	Real income per person in 2035 reaches 2.17 times its 2022 level in India and 1.25 times in Pakistan (model inputs).	**2026** urban affluent buyers. **2030** India's growth widens the gap. **2035** income alone leaves both far below US levels.	IMF revisions; household processed-food spending
Affordability	Healthy diet unaffordable for 40.4% (India) and 60.3% (Pakistan), 2024.	**2026** premium prices limit buying to trial. **2030** prices must approach chicken or paneer. **2035** compete with pulses and eggs.	Retail price per serving
Dietary practice	Pakistan: 32.6 g animal protein per day; India: 20.0 g. India's weekly meat, fish or chicken frequency among women ranges from 6.3% (Haryana) to 98.7% (Lakshadweep).	**2026** Indian sales lean on dairy alternatives; Pakistani products would target chicken and milk. **2030** fit existing meals. **2035** Pakistan has more animal food to replace.	Category sales; meal-based choice tests
Social legitimacy	Cultural variation gives no national willingness score. Within-country differences are large.	**2026** purposive samples only. **2030** check repeat purchase. **2035** community differences may exceed country differences.	Comparable bilateral purchase surveys
Certification	Halal conditions for cultivated meat are product-specific (Section 7).	**2026** ordinary halal assurance for plant-based foods. **2030** a certified cultivated product could set a precedent. **2035** fermentation certification becomes routine or stays contested.	Named Pakistani product decisions
Public policy	India: BioE3, FSSAI vegan and non-specified-food rules. Pakistan: halal institutions and a Punjab novel-food provision.	**2026** no cultivated authorisation in either country. **2030** fermentation foods may reach India first. **2035** cultivated meat stays excluded without approval.	Food approvals and notifications
Industry activity	Vacancies: India 529, Pakistan 58 (2025). Directory: 119 Indian and 2 Pakistani company rows.	**2026** India has brands and ingredient makers. **2030** continued Indian hiring could support the upper scenarios. **2035** Pakistan depends on a first local manufacturer.	Annual vacancies; Pakistani launches

AI is the eighth driver and has its own section. Sources: model inputs (World Bank, IMF WEO October 2026), SOFI 2025, NFHS-5, PBS HIES 2024 to 2025, AltProtein.Jobs 2023 to 2025.

3.4 Within-country variation

A national label hides large differences. In India's fifth National Family Health Survey, the share of women eating fish, chicken or meat at least weekly runs from 6.3% in Haryana and 7.0% in Punjab to 90.5% in Kerala. In Pakistan, 27.6% of women reached minimum dietary diversity in the 2018 to 2019 National Nutrition Survey, with provincial values from 16.5% (Sindh) to 40.3% (Khyber Pakhtunkhwa). The surveys measure different things, so the two countries cannot be ranked on one scale. The practical consequence is that a pilot in one city or province cannot stand for either nation.

4. AI as a driver

AI could lower costs for alternative proteins and for conventional animal production. Which effect wins is unresolved. The evidence supports a possibility and does not yet support an estimate.

4.1 What the evidence shows

Two peer-reviewed results anchor the mechanism. Cosenza and colleagues (2023) used Bayesian optimisation to find low-cost, high-growth serum-free culture media, tested in wet-lab cell lines. Watson and colleagues (2023) designed protein structures with RFdiffusion and confirmed them experimentally. Neither study measured food cost, safety or manufacturing economics, and neither tested a product in India or Pakistan.

Policy documents show intent. India's BioE3 policy, approved on 24 August 2024, provides for Bio-AI hubs and biofoundries. The 2025 DBT-BIRAC Smart Proteins call proposes AI tools for fermentation monitoring and predictive control. A research objective is not a grant award, and neither document shows a deployed system.

4.2 Company evidence is thin and undated

Table 2. AI claims by firm, with what each source can and cannot support

Firm	Country, sector	What the source states	What it does not show
Cell Forge	India, fermentation	Describes an AI-based platform for recombinant proteins and precision fermentation. Product section marked "coming soon."	Approval, sales, or any yield or cost gain from AI. No publication date.
Cowlar	Pakistan, dairy	Describes machine learning on cow temperature, activity and behaviour.	Current activity or farm coverage. Pages carry a 2017 footer.
Poulta	US/Canada, poultry; Pakistan trials	Advertises AI poultry optimisation. A programme case study points to Pakistani deployments from about 2021.	Current Pakistani operations or an isolated AI effect. The homepage lists no Pakistan address.

Dated observations, accessed 17 October 2026. A US Department of Agriculture report on precision dairy (ERR-356, 22 January 2026) supplies a US comparator, not a transferable coefficient.

4.3 Three speculative directions

- **2026.** Establish baselines. Record current AI use, yield and cost for named firms in both countries.
- **2030.** Indian alternative proteins and Pakistani dairy and poultry could each improve if independent pilots confirm gains. Both sectors would then gain at once.
- **2035.** Relative competitiveness depends on which sector reduces unit cost faster. A cheaper conventional chicken raises the bar for a plant-based substitute.

> **Why no AI uplift appears in the scenarios.** Any AI effect would overlap price, policy and industry activity, which the model already carries. Adding a separate multiplier would count the same change twice. A later quantified effect needs independently checked AI versus non-AI yield, cost per usable protein, and retail price gaps.

4.4 Why the incumbent matters

Conventional animal industries already hold data, capital and distribution. In Pakistan, poultry supply is 2.5 times India's, and dairy availability is 130 kg per person against 87 kg. If AI-led precision tools cut feed or disease losses first in these sectors, they lower the price of the foods that alternative proteins must beat. The same technology can therefore raise the threshold for entry. The September literature review frames AI as both an intervention and an incumbent advantage. Section 6 turns that framing into a funding rule.

4.5 What would change this assessment

- A dated, independent comparison of an Indian fermentation firm's yield or cost before and after AI tools.
- Farm-level Pakistani data that separate AI monitoring from sensors and management changes.
- Retail price gaps between plant-based and conventional chicken, tracked over at least two years.

The bounded English-language search on 17 October 2026 found none of these. The search is not a census of private deployments, so the absence is a gap in the record and does not show that no effect exists.

5. Market scenarios to 2035

Middle scenarios reach $448 million in India and $11.8 million in Pakistan by 2035. The ranges are wide, and the model sits below GFI India's published 2030 scenarios.

5.1 How the scenarios are built

India is anchored to GFI India's 2022 domestic estimate of 351 crore rupees, converted with inflation and the exchange rate to $45 million. Later values scale with real income, urbanisation and an annual drift for price, taste and availability. Pakistan has no domestic market observation. Its scenario scales India's by relative income, urban share, animal-protein supply and industry activity. IMF population data convert spending per person to totals. The inputs are analyst choices. The model has no calibration against verified sales.

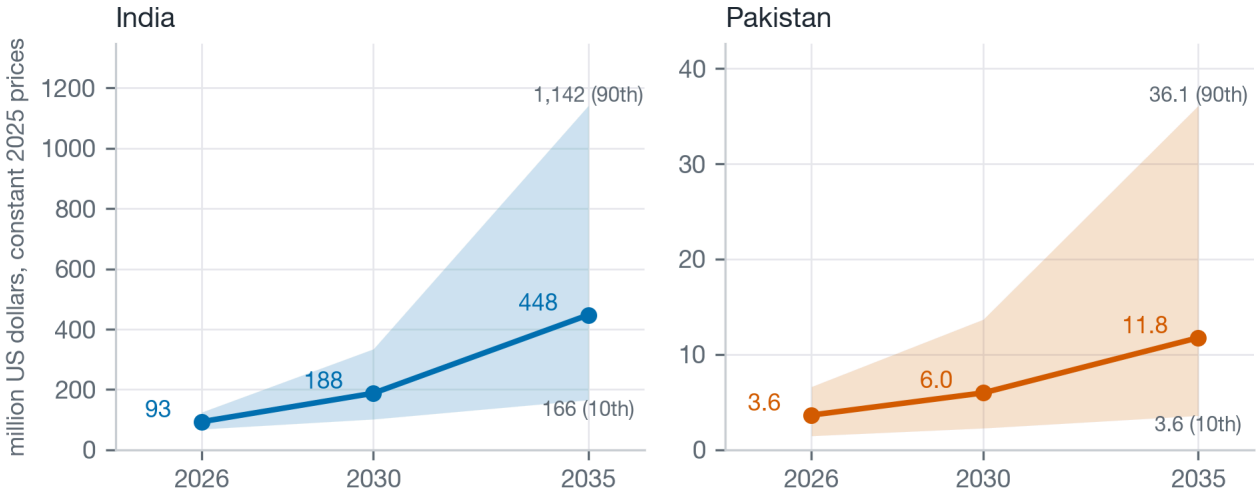


Figure 4. Spending grows fast from a small base and the range widens each year. Lines show the middle scenario (median of 10,000 draws). Shading spans the 10th to 90th percentile. Millions of US dollars, constant 2025 prices, market exchange rates. Panels use different scales.

Table 3. Scenario values (million US dollars, constant 2025 prices)

Country, year	10th	Middle	90th	Per person ($)	Share of US 2025
India 2026	69	93	125	0.06	0.27%
India 2030	102	188	334	0.12	0.53%
India 2035	166	448	1,142	0.28	1.22%
Pakistan 2026	1.5	3.6	6.6	0.015	0.06%
Pakistan 2030	2.3	6.0	13.7	0.023	0.10%
Pakistan 2035	3.6	11.8	36.1	0.040	0.17%

The US benchmark is GFI/SPINS plant-based retail sales of $7.9 billion in 2025, about $23.11 per person, and it includes prepared meals. Source: `altprotein_proxy_scenarios.csv` , derived tier.

5.2 Per-person spending stays near zero against the US

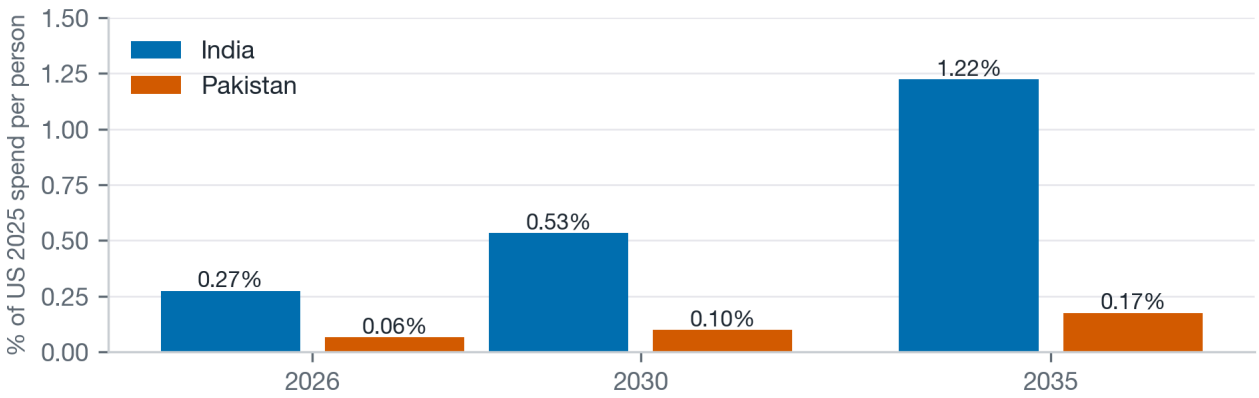


Figure 5. Even in 2035 India reaches about 1% of US spending per person. Middle scenario. Pakistan reaches 0.17%. The US figure is an observed 2025 value and not a forecast. Section 5.7 reports the separate US panel forecasts; this figure retains the observed 2025 comparison.

The result follows from low incomes, small existing product ranges and a thin base of commercial activity. It does not follow from low interest. The model contains no measure of interest.

5.3 The model sits below published India scenarios

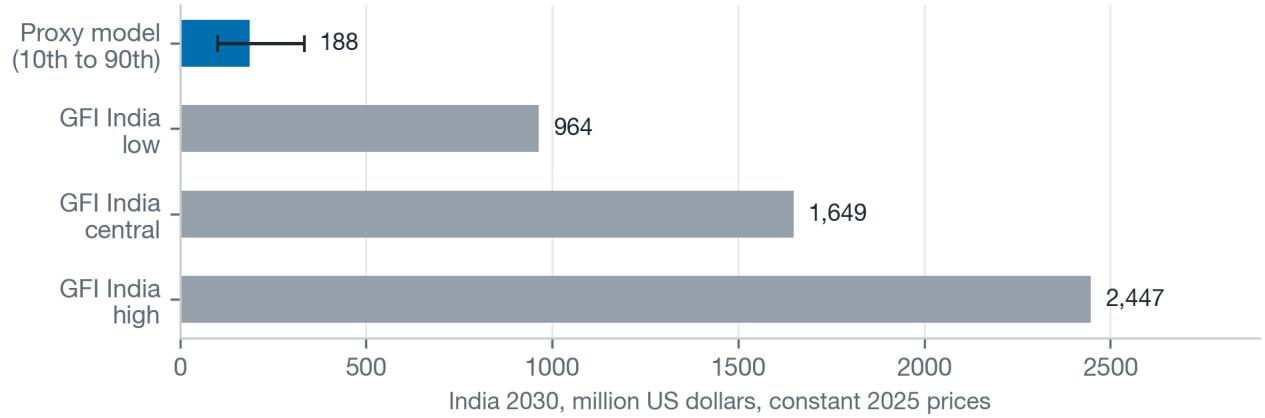


Figure 6. All 10,000 draws fall below GFI India's lowest 2030 scenario. GFI's low and central paths imply annual growth of 47% and 57%. The fastest draw grows 34% a year. Converted to the same dollar basis.

A reader will ask which side is wrong. The evidence cannot say. GFI India does not publish its model inputs, and it does not state whether its values are retail or nominal rupees. Both are assumed here. The proxy model's growth rests on income and urbanisation, which move slowly. GFI's scenarios may assume faster product launches and distribution. No test in hand separates the two explanations.

5.4 Two corrections that lowered the upper bound

The first version of the upper income elasticity (2.86) divided a market-rate spending gap by a purchasing-power income gap and mixed two years. Putting both on one basis and one year gives 1.81. The India 2035 upper bound fell from $1,945 million to $1,142 million. Second, India's high drift (17.8% a year) comes from job-vacancy growth, which already reflects some income growth. Removing the income share of that drift lowers the India 2035 middle value from $448 million to $215 million.

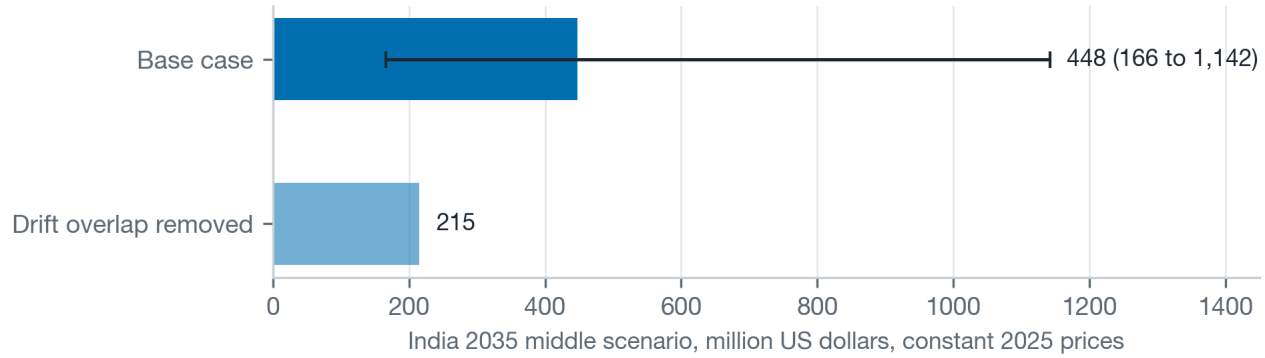


Figure 7. Possible double counting would halve India's 2035 middle value. Pakistan's four multipliers may also overlap. Removing its urban share and animal-protein factors gives a 2035 range of $2.6 million to $25.3 million.

5.5 The supply forecast is unreliable for India

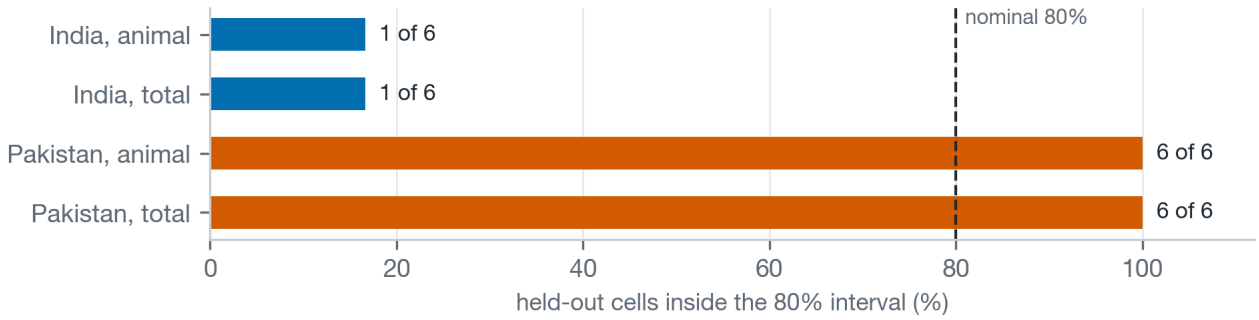


Figure 8. The nominal 80% interval covered 1 of 6 held-out India cells and 6 of 6 Pakistan cells. The evaluation set is small and its cells overlap. It supports no claim about future calibration. The benchmark forecasts national protein supply and not alternative-protein adoption.

5.6 How much weight each result carries

Table 4. Evidence tier and weight in this report's conclusions

Result	Source tier	Weight
Supply and diet differences	A: FAOSTAT, official household surveys	High for national availability. None for individual intake.
Affordability and stunting	A: SOFI and JME modelled estimates	Medium. Modelled, with different denominators.
Market scenarios	Derived: analyst-chosen inputs	Low. Speculative, with no calibration to sales.
Supply forecast	Derived from FAOSTAT history	Low for India (1 of 6 cells covered). Medium for Pakistan (6 of 6), on a small set.
Vacancies and directory	C: one job board	Low to medium. One platform, no role detail.
AI company claims	First-party web pages, undated	Low. Reported applications only.
Halal and policy instruments	Official documents and institutional statements	Medium for what a document says. None for its effect on purchases.
Consumer studies	Published, heterogeneous samples	Low for adoption. Stated intentions in sample-specific groups.

Tiers follow the research edition's source ledger. Weight is this report's judgement and awaits human review.

5.7 What an AI forecast panel adds

We adapted Anna Bartlett's US questionnaire for India and Pakistan; gpt-6-astra and claude-opus-5-5 supplied 10 runs per model and region, each in a fresh context. We retained 60 accepted runs from 79 attempts and report the median and 10th to 90th percentiles of the runs' own medians.

Anna agreed on 22 October 2026 to include all 60 runs, including the US runs, which also inform her US study. Table 3b reports each model separately for 2026, 2030 and 2035, in million US dollars, constant 2025 prices.

For India in 2035, gpt-6-astra gives 370 and claude-opus-5-5 gives 158 million US dollars, constant 2025 prices. For Pakistan, the corresponding medians are 37.0 and 9.35 million US dollars, constant 2025 prices. For the United States, they are 10,650 and 7,000 million US dollars, con-

stant 2025 prices. The two models' spreads do not overlap in any region by 2035. These spreads are the 10th to 90th percentiles of run medians, not calibrated uncertainty.

The separate proxy model puts India at 448 and Pakistan at 11.8 million US dollars, constant 2025 prices, in 2035. The proxy sits above both models for India and between them for Pakistan, so neither method brackets the other. We never average proxy scenarios with panel forecasts. Different inputs and methods make their comparison a prompt for checking assumptions.

This two-model panel has narrower model coverage than Anna's seven-model panel. Receipts configure gpt-6-astra with live web search, but do not establish executed searches. Receipts and the setup README enable WebSearch and WebFetch for claude-opus-5-5, but receipts do not establish executed searches. Repeated answers from one model do not constitute independent human forecasts. The panel describes disagreement; it does not measure calibrated uncertainty or validate either model against future sales. A shared questionnaire makes disagreement visible, but cannot establish which forecast will prove more accurate. Verified sales and later observations must supply that test.

Table 3b. AI forecast panel (million US dollars, constant 2025 prices)

Model	Region	Year	Median of run medians	10th to 90th percentiles of run medians
gpt-6-astra	India	2026	70.0	65.0 to 75.5
gpt-6-astra	India	2030	155	138 to 167
gpt-6-astra	India	2035	370	324 to 422
claude-opus-5-5	India	2026	50.0	30.9 to 65.0
claude-opus-5-5	India	2030	91.0	49.8 to 108
claude-opus-5-5	India	2035	158	84.2 to 191
gpt-6-astra	Pakistan	2026	6.00	5.00 to 7.20
gpt-6-astra	Pakistan	2030	14.5	10.9 to 18.4
gpt-6-astra	Pakistan	2035	37.0	27.6 to 60.0
claude-opus-5-5	Pakistan	2026	3.35	2.45 to 5.23
claude-opus-5-5	Pakistan	2030	5.50	4.44 to 9.01
claude-opus-5-5	Pakistan	2035	9.35	7.96 to 15.3
gpt-6-astra	United States	2026	7,600	7,550 to 7,650
gpt-6-astra	United States	2030	8,300	8,200 to 8,500
gpt-6-astra	United States	2035	10,650	10,500 to 11,000
claude-opus-5-5	United States	2026	7,600	7,595 to 7,655
claude-opus-5-5	United States	2030	7,200	7,090 to 7,300
claude-opus-5-5	United States	2035	7,000	6,880 to 7,320

Source: Edition 3 `data/forecasts.json`, `agg` rows keyed by model, region and year; `p50` and `p10` to `p90`, with `n=10`. Counts use `meta.accepted`, `meta.per_cell` and the sum of `meta.attempts`. Values under 1,000 use three significant figures with half-up rounding (157.5 becomes 158); values of 1,000 or more are shown in full. Unit: million US dollars, constant 2025 prices. The separate proxy

source is `research/india-pakistan/data/altprotein_proxy_scenarios.csv`, `country=India/Pakistan, year=2035, quantity=market_value, statistic=p50`. Search-access evidence: attempt `receipt.json` command fields and the elicitation README, line 7.

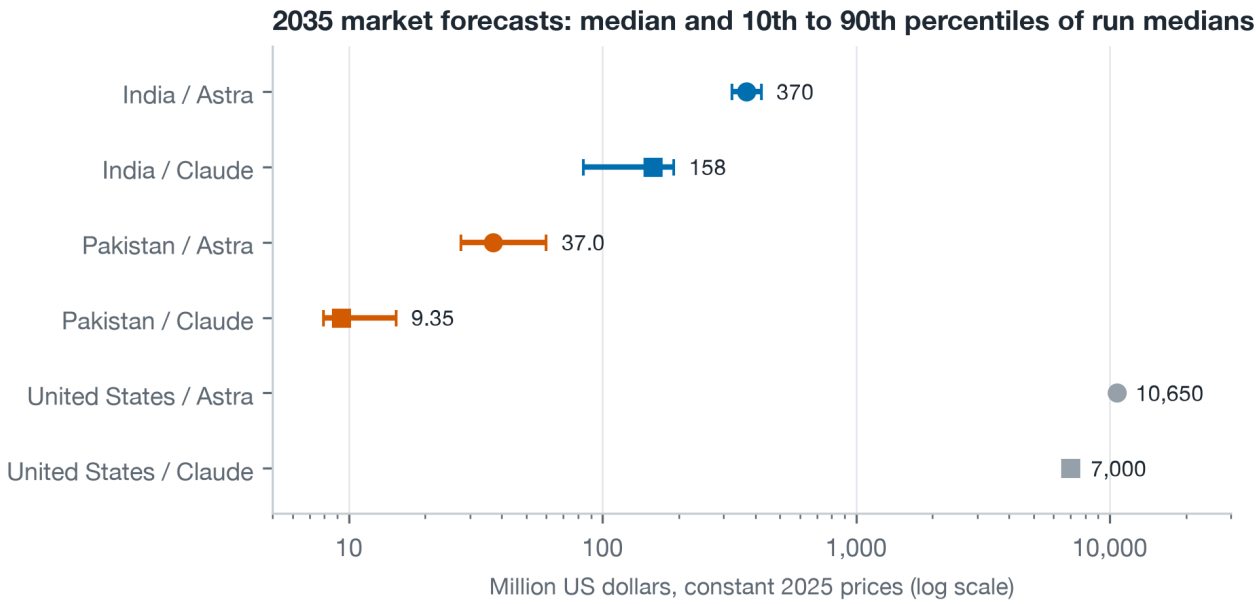


Figure 10. Two models give distinct 2035 forecasts in every region. Dots mark medians; horizontal bars show 10th to 90th percentiles of run medians. Circles represent gpt-6-astra; squares represent claude-opus-5-5. The axis uses a log scale; each major step multiplies value by ten. Source: forecasts.json, agg rows for 2035. These run spreads are not calibrated uncertainty.

6. Talent and resource allocation

Fund employer-linked technical trials in India and a small ingredient-application pilot in Pakistan before any broad training expansion. This is a proposal to test. It is not a ranking of returns.

6.1 What the job-board data show

Job postings measure industry demand only. The team plan asks about industry, research, policy and advocacy capacity. The evidence below broadens that assessment without estimating a national workforce.

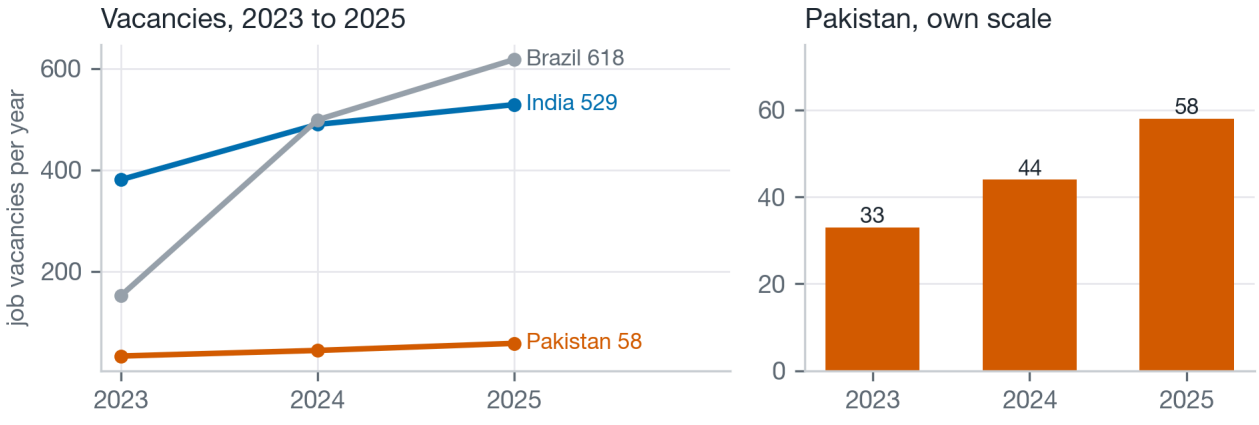


Figure 9. India's vacancies grew 39% over two years; Pakistan's grew 76% from a base of 33. India posted 529 vacancies in 2025, Pakistan 58, Brazil 618. Brazil overtook India in 2024. Source: AltProtein.Jobs annual workbook 2023 to 2025.

Table 5. Talent indicators

Country	2023	2024	2025	Per million people, 2025	Directory rows
India	381	490	529	0.36	119
Pakistan	33	44	58	0.24	2
Brazil	152	499	618	not computed	46

Vacancies are job postings on one platform. Directory rows are listed companies and not annual counts. Per-person rates use IMF population, October 2026.

Per million people the gap narrows from a factor of 9 in raw vacancies to a factor of 1.5. Vacancies measure advertised demand on one job board. They do not measure employment, successful hiring or skill shortages. The workbook has no role or function column, so it cannot show which skills employers want.

6.2 Proposed allocation

Table 6. Allocation proposals and how each would be tested

Where	Proposal	Test before expansion
India	Employer-linked technical trials. Ask employers to name one technical constraint, then compare training with access to shared facilities.	Time from award to validated result, measured against a baseline for each employer.
Pakistan	A small ingredient-application pilot with plant-based ingredient firms. Shafi Gluco Chem has two overlapping directory entries, so facilities and staff need confirming first.	Reproducible ingredient specifications, safety documentation and a usable formulation.
Computational talent	Fund only where evidence shows computation, data or experimental design limits progress. Pair every award with experimental validation.	A laboratory gain does not show lower retail prices. Track price per usable protein as well.

Neither the vacancy workbook nor the AI literature review measures time to hire or time to build capacity, so the first grants must create that measure. Vacancies are not added to the market scenarios as another growth multiplier. The proxy already uses vacancy growth as drift, and a second use would double-count it.

6.3 Capacity beyond job postings

Capacity includes training, research infrastructure, regulatory expertise and organisations that connect these activities. The India documents describe parts of this system without providing a national workforce census. We do not combine their observations, recommendations and organisational descriptions into a capacity total.

GFI India's talent study gathered 20 industry responses from startups and larger companies, alongside a cohort of 19 academic leaders and administrators. These are participants, not national employer or worker counts. Among academic respondents, 47.4% reported adequate coverage

of smart protein topics in their curricula. This small study does not establish national representativeness. The document does not state a publication date.

The saved GFI company database explains its global coverage of manufacturers and suppliers. It contains no extractable India company rows and cannot supply an India capacity count. Curriculum coverage cannot establish laboratory access, practical competence or successful hiring.

The cultivated meat action plan recommends shared work on hazard identification, toxicology and risk assessment. These are recommended capabilities, not measured staffing or completed programmes. GFI India describes science, business and public sector engagement, evidence of an advocacy and convening role rather than its scale. Green Queen reports a proposed national training consortium linking skilling councils, industry and research institutions. It covers the same GFI study and does not independently confirm an established consortium.

No equivalent Pakistan source exists on disk. Missing evidence does not establish missing capacity, and we infer no Pakistan values. Comparison needs matched evidence on institutions, staff, facilities and completed training. No talent figures from this section enter the market scenarios.

Table 5b. Capacity evidence beyond job postings

Category	India evidence	Pakistan evidence	Source
Industry	20 study responses; global directory supplies no extractable India company rows.	Not available locally.	TAL-01, p. 10; TAL-04, p. 2
Research	19 academic respondents; 47.4% reported adequate curriculum coverage.	Not available locally.	TAL-02, p. 10; TAL-03, p. 16
Policy	Proposed shared hazard, toxicology and risk assessment.	Not available locally.	TAL-05, p. 7
Advocacy	GFI convening role; proposed consortium, reported by Green Queen.	Not available locally.	TAL-06, p. 10; TAL-07, p. 6

7. Certification and policy conditions

Religious approval, food authorisation and market uptake are three separate events. A position on one does not settle the others.

7.1 Halal conditions for cultivated chicken

The International Islamic Fiqh Academy's 2025 resolution makes its cultivated-meat position conditional on the donor, culture inputs and oversight. It excludes prohibited culture media, requires supervision by a credible specialist authority, and assigns disclosure and verification duties. Singapore's Islamic Religious Council (MUIS) stated in 2024 that the cell source must be a halal animal and that texture and composition ingredients must be halal. Neither statement certifies a particular manufacturer's product, and neither grants food authorisation in Pakistan.

The cultivated-chicken halal paper separates six questions: donor procurement, culture inputs, transformation, category and edibility, certification, and food authorisation. Its source register holds 95 records. Of these, 63 were verified with a note by the recorded checking process, 22 are held and 10 were not retrieved. All 95 await human verification, so "verified with a note" is a source-audit status and not scholarly approval.

7.2 Policy instruments in both countries

Table 7. Selected instruments, with what each does and does not establish

Country	Instrument and date	What it establishes
India	BioE3 policy, approved 24 Aug 2024	Innovation policy with Bio-AI hubs and biofoundries. It does not approve a food product.
India	DBT-BIRAC Smart Proteins call, 22 Feb 2025	A closed funding call. A funded project is not a commercial product.
India	FSSAI non-specified-food compendium, version 14 Oct 2022	A general prior-approval route for novel foods.
India	Vegan-food regulations (2022) and later amendments (commence 2027)	Labelling framework. Publication dates differ from commencement dates.
India	Biokraft application, filed 20 Aug 2025, listed 1 Sep 2026	An application stage with ambiguous status. No approval is inferred.
Pakistan	Punjab functional-food regulations, notified 27 Nov 2025, gazetted 4 Sep 2026	A general novel-food provision exists. No cultivated-specific approval route is documented.
Pakistan	Pakistan Halal Authority mark scheme, operationalised 20 May 2025	A general halal certification scheme. It does not approve a named product.

Full instrument register with 15 dated entries: research edition, chapter 10 and policy appendix.

The comparison is not India having policy and Pakistan having none. India has research funding and food-regulatory instruments. Pakistan has halal institutions and one provincial novel-food provision. Neither country has an identified approval for a named cultivated product. The practical question is which documents a specified product would need and whether the responsible institutions have issued them.

7.3 Implication for scenarios

The scenarios carry no cultivated meat. If a certified cultivated product reached Pakistan by 2030, it would set a precedent for fermentation-derived foods too. That event is one of the signposts in Table 1. The halal workstream's quantitative phase is blocked until compatible demand inputs exist.

8. What the scenarios do not measure, and what to do next

8.1 Four outcomes that stay separate

The scenarios estimate retail spending. They do not count households trying, repeating or switching. Four outcomes need separate evidence: stated willingness to buy, first purchase, repeat purchase after incentives end, and a measured fall in animal-food purchases. Spending growth cannot tell one-time buyers from repeat buyers.

8.2 The consumer evidence

The search found nine study records and 29 findings, with no India finding. Bryant and colleagues (2019) measure hypothetical purchase intentions, including 1,024 Indian respondents. Arora, Brent and Jaenicke (2020) measure stated choices. Ahsan, Uzair and Ali (2021) measure Pakistani attitudes, and their acceptance reporting is inconsistent. Dean and colleagues (2023) measure pooled willingness to try, buy and pay more, including Pakistan. These are sample-specific responses. They give no national purchase rate and no observed first purchase, repeat purchase or substitution in either country.

Table 8. The four bilateral consumer studies and what each measures

Study	Coverage	What it measures	What it cannot show
Bryant et al. (2019)	USA, India, China; 1,024 Indian respondents	Hypothetical purchase intentions	Actual purchase or repeat use
Arora, Brent and Jaenicke (2020)	Consumer sample in India	Stated choices among options	Observed purchase at a shelf
Ahsan, Uzair and Ali (2021)	Pakistan	Attitudes and perceptions	A reliable acceptance rate (reporting is inconsistent)
Dean et al. (2023)	Several countries including Pakistan	Pooled willingness to try, buy and pay more	A country-specific rate

Register: 9 studies and 29 findings, with no India finding. Results are not pooled across countries.

8.3 Three studies that would test household demand

Table 9. Proposed studies. None is an observed result.

Study	Design	Result that would count against the mechanism
Entry cost	Randomise pack size and unit price independently. Record cash outlay, price per meal, household budget and actual first purchase.	No pack-outlay effect once unit price is controlled, with uncertainty narrow enough to exclude a useful effect.
Household meal fit	Randomise meal-use guidance. Collect meal diaries and dated purchase receipts after incentives end.	Similar unsubsidised repeat purchase across groups despite a verified difference in shared-meal use.
Replacement	Follow household purchases before trial and after incentives end, including non-triers. Record chicken, pulses and all competing foods.	Unchanged animal-food purchases alongside repeat alternative-protein purchases, or replacement of pulses alone.

The purchase-tracking design lasts 12 weeks. Its primary window is 28 days, followed by a common 56-day repeat window for buyers observed by day 28. First observed paid purchase among recent non-buyers is not verified first-ever trial. The design reports low conversion as readily as high conversion. No provider, sample, budget or field start has been chosen, and an access request is drafted but unsent.

A 16 October 2026 retail capture holds 97 verified prices from one outlet per country, Modern Bazaar in India and Naheed.pk in Pakistan. It holds no verified protein values and no plant-based meat category. Price comparisons wait on more outlets and that category.

8.4 Recommendations

1. **Use the shared definitions.** Agreed with the US and Africa authors on 22 October 2026: US dollars at market rates, constant 2025 prices, retail value, and the product scope in Table A1. The US anchor includes prepared meals, which the scope does not name; every US comparison states this.
2. **Replace the proxy with a sales observation.** Obtain verified domestic sales for at least one Indian category. That does more for the scenarios than refining the model.
3. **Run the purchase-tracking study.** Link stated intention to verified transactions in a defined regional cohort. Report the conversion rate whatever it is.
4. **Fund the allocation trials in Section 6.** Measure time to validated result before scaling either programme.
5. **Gate AI claims.** Accept an AI effect into a scenario only with independently checked yield, cost-per-protein and retail-price data.
6. **Monitor the signposts.** Review Table 1's signposts every six months: IMF revisions, price per serving, approvals, vacancies and Pakistani launches.

Open items before submission. (1) Human scholarly review of all numbers. (2) One baseline mismatch in the data package: the 2023 animal-protein supply used as a model input (19.67 and 31.52 g per person per day) differs from the benchmark series shown in Figure 1 (20.01 and 32.55 g). It needs reconciling at the source.

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Appendix A. Definitions and open items

Table A1. Shared definitions, agreed with the US and Africa authors on 22 October 2026

Field	Agreed definition
Currency	US dollars at market exchange rates
Price year	Constant 2025 prices
Value basis	Retail (consumer) sales
Product scope	Plant-based meat, dairy and egg alternatives and fermentation-derived foods sold for domestic human consumption
Comparison metric	Spending per person as a share of US spending per person
Forecast years	2026, 2030, 2035

Known mismatch: the US anchor includes prepared meals, which the scope does not name.

Table A2. Evidence limits that stay open

Gap	Status on 20 October 2026
Pakistan whole-diet protein	Withdrawn. Named physical conversions cover subsets only: five animal foods give 3.30 to 4.64 g per person per day and a matched basket 39.27 to 40.61 g. These are extrema of named cases and not confidence intervals.
Category crosswalks (US to India)	31 reviewed: 21 partial, 5 incompatible, 5 unknown. None exact or admitted for recalibration.
Retail price audit	22 city and category cells dispositioned. Two limited alternative-dairy listings evidenced. No Pakistan meat price admitted.
Official production rules	Not found in fresh official sources. Nine precise questions and two unsent drafts prepared.
Pakistan report discrepancy	Two report URLs return identical bytes. The workbook and report disagreement persists.
Human review	Pending for every section. Source checking and export integrity do not constitute scholarly approval.

Appendix B. Figure and table data sources

Item	Data file in the supervisor edition package (13 October 2026)
Figure 1	`benchmark_annual.csv` (FAOSTAT, tier A)
Figure 2	`meat_supply_national.csv`, metric food_supply_quantity, 2023
Figure 3	Edition chapter 8 indicators, SOFI 2025 and JME 2026 (tier A)
Figures 4 to 7, Table 3	`altprotein_proxy_scenarios.csv` (derived), seed 20261015, 10,000 draws; `tools/build_altprotein_proxy.py`
Figure 8	`benchmark_coverage.csv` (derived)
Figure 10, Table 3b	`data/forecasts.json`, Edition 3: `agg` rows keyed by model, region, year; `p10`, `p50`, `p90`. Counts: `meta`. Three significant figures, half-up rounding.
Figure 9, Table 5	`altprotein_jobs.csv` (tier C, job-board)